

Homework #2

Logic

1. Analyze the logical forms of the following statements.

- (a) Either John and Bill are both telling the truth, or neither of them is.

let:

$J = \text{John}$

$B = \text{Bill}$

The logical form can be written out as: $(J \wedge B) \vee (\neg J \wedge \neg B)$

- (b) I'll have either fish or chicken, but I won't have both fish and mashed potatoes.

let:

$F = \text{Fish}$

$C = \text{Chicken}$

$M = \text{Mashed Potatoes}$

The logical form can be written out as: $(F \vee C) \wedge \neg(F \wedge M)$

- (c) 3 is a common divisor of 6, 9, and 15

The logical form can be written out as: $6, 9, 15 \in \{x | x \% 3 = 0\}$

2. Which of the following expressions are well-formed formulas?

- (a) $\neg(\neg P \wedge \neg \neg R)$, is well formed. While the double $\neg$ is somewhat irksome its still correct
- (b) $\neg(P, Q, \vee R)$ No, as the commas make it not proper "Syntax"
- (c) $P \wedge \neg P$ Well formed, as it is clear and not ambiguous. Unless parenthesis are required.
- (d) $(P \wedge Q)(P \vee R)$ Not well formed.

3. Let T stand for the statement "Taxes will go up" and D for "The deficit will go up." What English sentences are represented by the following formulas?

- (a) $T \vee D$: Taxes will go up or The deficit will go up

- (b) $(T \wedge D) \wedge \neg(\neg T \wedge \neg D)$:

Taxes and Decifitit will go up and Taxes and Deficit will both not not go up.

- (c) $(T \wedge \neg D) \vee (D \wedge \neg T)$:

Taxes and will go up and Deficit won't or Deficit will go up and Taxes wont.

4. Identify the premises and conclusions of the following deductive arguments and analyze their logical forms. Do you think the reasoning is valid? (Although you will have only your intuition to guide you in answering this last question, in the next section we will develop some techniques for determining the validity of arguments.)

- (a) The main course will be either beef or fish. The vegetable will be either peas or corn. We will not have both fish as a main course and corn as a vegetable. Therefore, we will not have both beef as a main course and peas as a vegetable.

False, as Beef and Peas are unconstrained and are able to be a pair without a contradiction on the Fish and Corn constraint

- (b) Either John or Bill is telling the truth. Either Sam or Bill is lying. Therefore, either John is telling the truth or Sam is lying.

This is true, if John is true then it Makes Bill false and forces Sam to true, and When John is false, a similar chain of reasoning makes Sam False

1.2 Exercises

5. Make truth tables for the following formulas:

- (a) $\neg[P \wedge (Q \vee \neg P)]$

P	Q	$\neg P$	$Q \vee \neg P$	$P \wedge (Q \vee \neg P)$	$\neg[P \wedge (Q \vee \neg P)]$
T	T	F	T	T	F
T	F	F	F	F	T
F	T	T	T	F	T
F	F	T	T	F	T

- (b) $(P \vee Q) \wedge (\neg P \vee R)$

P	Q	R	$P \vee Q$	$\neg P \vee R$	$(P \vee Q) \wedge (\neg P \vee R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	F	T	F
F	F	F	F	T	F

6. In this exercise we will use the symbol $+$ to mean exclusive or. In other words, $P + Q$ means "P or Q, but not both."

- (a) Make a truth table for $P + Q$.

P	Q	$P + Q$
T	T	F
T	F	T
F	T	T
F	F	F

- (b) Find a formula using only the connectives $\wedge$, $\vee$, and $\neg$ that is equivalent to $P + Q$. Justify your answer with a truth table.

P	Q	$P + Q$	$\neg Q$	$P \wedge \neg Q$	$\neg P \wedge Q$
T	T	F	F	F	F
T	F	T	T	T	F
F	T	T	F	F	T
F	F	F	T	F	F

$(P \wedge \neg Q) \vee (\neg P \wedge Q)$		$P + Q$
F		F
T		T
T		T
F		F

7. Find a formula using only the connectives $\wedge$ and $\neg$ that is equivalent to $P \vee Q$. Justify your answer with a truth table.

(a) when can make it with demorgans law

P	Q	$P \vee Q$	$\neg P$	$\neg Q$	$\neg(\neg P \wedge \neg Q)$
T	T	T	F	F	T
T	F	T	F	T	T
F	T	T	T	F	T
F	F	F	T	T	F

8. Some mathematicians write $P \mid Q$ to mean “P and Q are not both true.” (This connective is called nand, and is used in the study of circuits in computer science.)

(a)

P	Q	$P \wedge Q$	$P \mid Q = \neg(P \wedge Q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

9. Use truth tables to determine which of the following formulas are equivalent to each other

(a) $(P \wedge Q) \wedge (\neg P \vee \neg Q)$

P	Q	$\neg P$	$\neg Q$	$P \vee Q$	$\neg P \vee \neg Q$	$(P \vee Q) \wedge (\neg P \vee \neg Q)$
T	T	F	F	T	F	F
T	F	F	T	T	T	T
F	T	T	F	T	T	T
F	F	T	T	F	T	F

(b) $\neg P \vee Q$

P	Q	$\neg P$	$\neg P \vee Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

(c) $(P \vee \neg Q) \wedge (Q \vee \neg P)$

P	Q	$\neg P$	$\neg Q$	$P \vee \neg Q$	$Q \vee \neg P$	$(P \vee \neg Q) \wedge (Q \vee \neg P)$
T	T	F	F	T	T	T
T	F	F	T	T	F	F
F	T	T	F	F	T	F
F	F	T	T	T	T	T

(d) $\neg(P \vee Q)$

P	Q	$P \vee Q$	$\neg(P \vee Q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

(e) $(Q \wedge P) \vee \neg P$

P	Q	$Q \wedge P$	$\neg P$	$(Q \wedge P) \vee \neg P$
T	T	T	F	T
T	F	F	F	F
F	T	F	T	T
F	F	F	T	T

(A) and (C) are equal and (B) and (E) and (D) is unique